Question and design

Does model width improve contact accuracy at the same one-epoch exposure? Baseline: exp277, native plus MPNN corpus, 24 layers. Hidden width 2048 -> 4096; MLP width 8192 -> 16384. Keep 32 query / 8 KV heads; head width 64 -> 128. Keep data, optimizer schedule, seeds and global batch fixed.

Architecture and smoke verified

5,861,984,256 parameters, versus exp277's 1,471,374,336 (3.984x). Architecture, tokenizer, data and recovery checks passed. 64-H100 smoke: ten updates, train loss 8.56948, eval loss 8.03575. 2.821 seconds/update; native and HF checkpoints saved with tokenizer. Compressed smoke schedule was spiky; this is an execution check.

Current status — October 19

GPU production running: step 48,451, 18.19% of one epoch. Train loss 2.893; validation loss 3.123, previously 3.465. Complete rolling checkpoint step 48,452; permanent step 26,634. 32 H100s: 5.61 s/update; about 14.2 compute days remaining. Queueing, validation, checkpoints and preemptions add wall time.

TPU trial succeeded

Interactive v5p-64: 32 chips across 8 VMs; all 20 updates completed. Finished October 16; no worker failures or preemptions. Native and HF step-19 checkpoints verified; tokenizer included. Median steps 5–19: 7.40 s/update, about 32% longer than GPU steps. Short native-only smoke; throughput is preliminary, not cost-normalized. Production continues on GPUs. No bulk transfer or migration performed.